MATH60005/70005: Optimisation
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Chapter 2: Linear Least Squares

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This week we explore a very important class of problems with different applications in maths, science, and engineering. Least Squares Problems are everywhere, they arise every time we want to fit a model to match observations/data. After a brief historical presentation, we will formulate the classical linear least squares problem and its solution through normal equations. We will explore connections to data fitting, and introduce a regularization term to cast an optimization problem that is central in signal denoising applications. We will further discuss the formulation of nonlinear least squares problems, and conclude with the case study of fitting data to a circle.

A Bit of History

In January 1, 1801, the Italian monk Giuseppe Piazzi, discovered a faint, nomadic object through his telescope in Palermo, correctly believing it to reside in the orbital region between Mars and Jupiter. Piazzi watched the object for 41 days but then fell ill, and shortly thereafter the wandering star strayed into the halo of the Sun and was lost to observation. The newly-discovered planet had been lost, and astronomers had a mere 41 days of observation covering a tiny arc of the night from which to attempt to compute an orbit and find the planet again.

Figure 1: Giusepe Piazzi (1746-1826), Italian priest, mathematician, and astronomer. His observations led to the discovery of the dwarf planet Ceres.

The dean of the French astrophysical establishment, Pierre-Simon Laplace, declared that the orbit recovery simply could not be done. In Germany, the 24 years old German mathematician Car Friedrich Gauss had considered that this type of problem, to determine a planet's orbit from a limited handful of observations- "commended itself to mathematicians by its difficulty and elegance." Gauss discovered a method for computing the planet's orbit using only three of the original observations and successfully predicted where Ceres might be found (now considered to be a dwarf planet). The prediction catapulted him to worldwide acclaim.

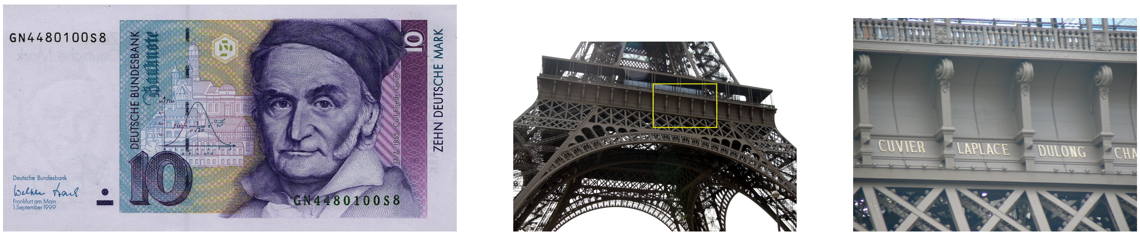

Figure 2: Carl Friedrich Gauss (1777-1855) and Pierre-Simon Laplace (1749-1827) need no introduction. On the left, a 10 Deutsche Mark bank note with Gauss. Laplace has his name engrave in the Eiffel Tower. How cool is that!

More than 200 years later, in 2019 American computer scientist Katie Bouman used similar mathematical methods, which heavily rely on optimization and large-scale astronomical observation datasets, to recover the first image of a black hole.

Figure 3: 200 years later, Katie Bouman (1990-) used similar optimization techniques as Gauss (and a bit of supercomputing power) to recover the first image of a black hole. After this module, you could be next!

Formulating the Linear Least Squares Problem

Consider the linear system

$$\mathbf{S}\mathbf{x} \approx \mathbf{b}, \qquad \left(\mathbf{S} \in \mathbb{R}^{m\times n}, \mathbf{b} \in \mathbb{R}^{m}\right)$$

We assume that S has a full column rank, that is, $rank(\mathbf{S}) = n$. However, when $m > n$, that is when you have more equations than unknowns, the system is usually inconsistent and a common approach for finding an approximate solution is to pick the solution of the problem

$$\min_{\mathbf{x}} \|\mathbf{S}\mathbf{x} - \mathbf{b}\|^2 , \qquad \text{(LLS)}$$

which is the same as (check!)

$$\min_{\mathbf{x}\in\mathbb{R}^n} \left\{ f(\mathbf{x}) \equiv \mathbf{x}^\top \mathbf{S}^\top \mathbf{S}\mathbf{x} - 2\mathbf{b}^\top \mathbf{S}\mathbf{x} + \|\mathbf{b}\|^2 \right\}$$

Note that $\nabla^2 f(\mathbf{x}) = 2\mathbf{S}^\top\mathbf{S} \succ 0$ since $rank(\mathbf{S}) = n$ and $m > n$. Therefore, the unique optimal solution $\mathbf{x}_{LS}$ is the solution $\nabla f(\mathbf{x}) = 0$, namely,

$$\left(\mathbf{S}^\top\mathbf{S}\right)\mathbf{x}_{\text{LS}} = \mathbf{S}^\top\mathbf{b} \leftarrow \text{ Normal Equations}\,,$$

leading to

$$\mathbf{x}_{\text{LS}} = \left(\mathbf{S}^\top\mathbf{S}\right)^{-1}\mathbf{S}^\top\mathbf{b}\,.$$

A Numerical Example. Consider the inconsistent linear system

$$\begin{aligned} x_1 + 2x_2 &= 0 \\ 2x_1 + x_2 &= 1 \\ 3x_1 + 2x_2 &= 1 \end{aligned}$$

To find the least squares solution, we will solve the normal equations:

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 2 \end{pmatrix}^\top \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 2 \end{pmatrix}^\top \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

which is the same as

$$\begin{pmatrix} 14 & 10 \\ 10 & 9 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} \Rightarrow \mathbf{x}_{LS} = \begin{pmatrix} 15/26 \\ -8/26 \end{pmatrix}.$$

Note that $\mathbf{A}\mathbf{x}_{LS} = (-0.038; 0.846; 1.115)$, so that the errors are

$$\mathbf{b} - \mathbf{S}\mathbf{x}_{LS} = \begin{pmatrix} 0.038 \\ 0.154 \\ -0.115 \end{pmatrix} \Rightarrow \text{ sq. error } = 0.038^2 + 0.154^2 + (-0.115)^2 = 0.038\,.$$

What is the core numerical task for solving LLS?

Data Fitting

We revisit, from an optimization viewpoint, a fundamental problem in statistics, namely, linear regression. Assume we have a dataset $(s_i, b_i)\,, i = 1, 2, \ldots, m$, where $s_i \in \mathbb{R}^n$ and $b_i \in \mathbb{R}$. Assume that an approximate linear relation holds:

$$b_i \approx \mathbf{s}_i^\top \mathbf{x}, \quad i = 1, 2, \ldots, m\,.$$

The corresponding least squares/regression problem reads

$$\min_{\mathbf{x} \in \mathbb{R}^n} \sum_{i=1}^{m} \left(\mathbf{s}_i^\top \mathbf{x} - b_i\right)^2,$$

or equivalently

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{S}\mathbf{x} - \mathbf{b}\|^2\,.$$

where

$$S = \begin{pmatrix} s_1^\top \\ s_2^\top \\ \vdots \\ s_m^\top \end{pmatrix}, \quad t = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

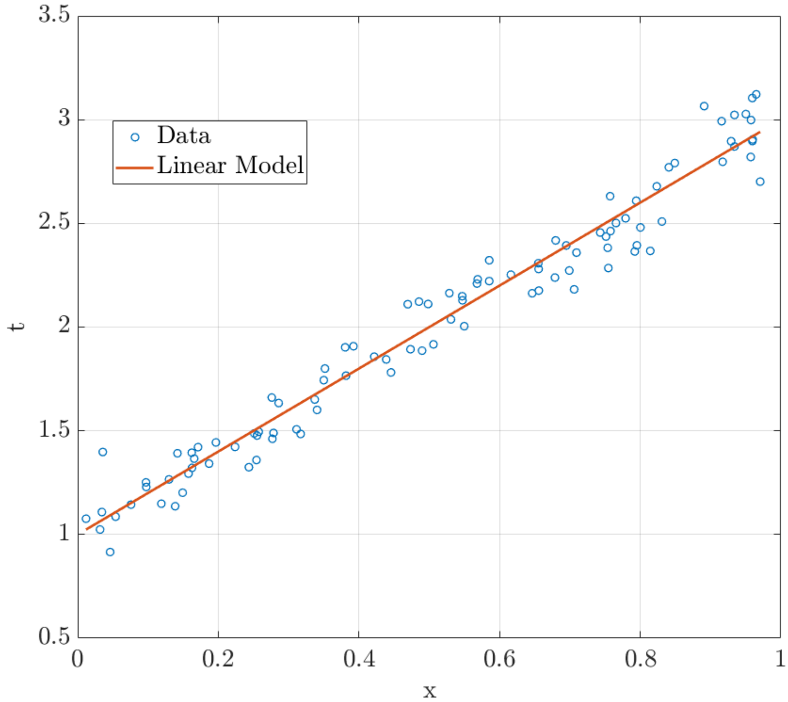

Figure 4: The typical situation in linear regression. A set of scattered data (s_i, t_i) (in blue) suggests a linear relation between s and t. Among all the possible linear models, there is an optimal choice (in red) which minimizes the square error between the model and the measurements.

Polynomial Fitting

Another problem relevant in statistics that can be cast as a linear least squares problems is the polynomial fit of data. Given a set of points in $\mathbb{R}^2 : (u_i, y_i), i = 1, 2, \ldots, m$, for which the following approximate relation holds for some $a_0, \ldots, a_d$:

$$\sum_{j=0}^{d} a_j u_i^j \approx y_i, \quad i = 1, \ldots, m$$

The linear system associated to the relation reads:

$$\underbrace{\begin{pmatrix} 1 & u_1 & u_1^2 & \cdots & u_1^d \\ 1 & u_2 & u_2^2 & \cdots & u_2^d \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & u_m & u_m^2 & \cdots & u_m^d \end{pmatrix}}_{\mathbf{U}} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_d \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_m \end{pmatrix}$$

The least squares solution is of course well defined if the $m \times (d+1)$ matrix is of full column rank. This is true when all the u_i 's are different from each other (why?).

Regularized Least Squares

There are several situations in which the least squares solution does not give rise to a good estimate of the "true" vector $\mathbf{x}$. In these cases, a regularized problem (called regularized least squares (RLS)) is often solved:

$$\textbf{(RLS)} \qquad \min_{\mathbf{x}} \|\mathbf{Sx} - \mathbf{b}\|^2 + \lambda R(\mathbf{x}) \,.$$

Here, λ is the regularization parameter and $R(\cdot)$ is the regularization function (also called a penalty function). A common choice is a quadratic regularization function:

$$\min \|\mathbf{Sx} - \mathbf{b}\|^2 + \lambda\|\mathbf{Dx}\|^2 \,.$$

The optimal solution of the above problem is (why?)

$$\mathbf{x}_{\mathrm{RLS}} = \left(\mathbf{S}^\top \mathbf{S} + \lambda \mathbf{D}^\top \mathbf{D}\right)^{-1} \mathbf{S}^\top \mathbf{b} \,.$$

What kind of assumptions are needed to assure that $\mathbf{S}^\top \mathbf{S} + \lambda \mathbf{D}^\top \mathbf{D}$ is invertible? (answer: $\mathrm{Null}(\mathbf{S}) \cap \mathrm{Null}(\mathbf{D}) = \{0\}$)[1].

Denoising

A very important application of linear least squares and regularization techniques is the denoising of signals (acoustic, images). Suppose that a noisy measurement of a signal $\mathbf{x} \in \mathbb{R}^n$ is given:

$$\mathbf{b} = \mathbf{x} + \mathbf{w} \,,$$

where $\mathbf{x}$ is the “true” unknown signal, $\mathbf{w}$ is the unknown noise and $\mathbf{b}$ is the (known) measures vector. Note that the least squares problem:

$$\min_{\mathbf{x}} \|\mathbf{x} - \mathbf{b}\|^2 \,,$$

is meaningless, as we would trivially recover $\mathbf{x} = \mathbf{b}$, without any denoising. We need to enrich the optimization problem by adding a suitable regularization term, exploiting some a priori information of the signal. For example, if the we know that the signal is “smooth" in some sense, then $R(\cdot)$ can be chosen as a penalization of the signal “sudden variations”

$$R(\mathbf{x}) = \sum_{i=1}^{n-1} (x_i - x_{i+1})^2 \;,$$

[1] here $\mathrm{Null}(\mathbf{S})$ refers to the nullspace of the matrix or kernel, $\mathrm{Ker}(\mathbf{S}) := \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{Sx} = \mathbf{0}\}$.

where the regularization $R(\cdot)$ can also be written as $R(\mathbf{x}) = \|\mathbf{L}\mathbf{x}\|^2$ where $\mathbf{L} \in \mathbb{R}^{(n-1)\times n}$ is given by

$$\mathbf{L} = \begin{pmatrix} 1 & -1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & -1 \end{pmatrix} .$$

The resulting regularized least squares problem is

$$\min_{\mathbf{x}} \underbrace{\|x - b\|^2}_{\text{fitting}} + \lambda \underbrace{\|Lx\|^2}_{\text{denoisig}} .$$

The direct solution of this problem leads to (why?)

$$\mathbf{x}_{\mathbf{RLS}}(\lambda) = \left(\mathbf{I} + \lambda \mathbf{L}^\top \mathbf{L}\right)^{-1} \mathbf{b} .$$

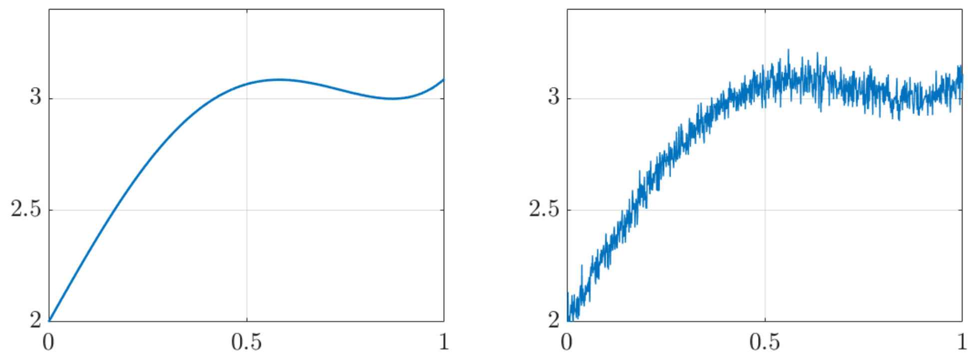

Figure 5: Signal denoising. We only have access to the noisy signal (left), and we would like to recover a "clear" signal (right) by solving a regularized least squares problem.

Nonlinear Least Squares

The least squares problem $\min \|\mathbf{S}\mathbf{x} - \mathbf{b}\|^2$ is often called linear least squares. In some applications we are given a set of nonlinear equations:

$$f_i(\mathbf{x}) \approx b_i, \quad i = 1, 2, \ldots, m .$$

The nonlinear least squares (NLS) problem is the one of finding an $\mathbf{x}$ solving the problem

$$\min_{\mathbf{x}} \sum_{i=1}^{m} \left(f_i(\mathbf{x}) - b_i\right)^2 . \tag{NLS}$$

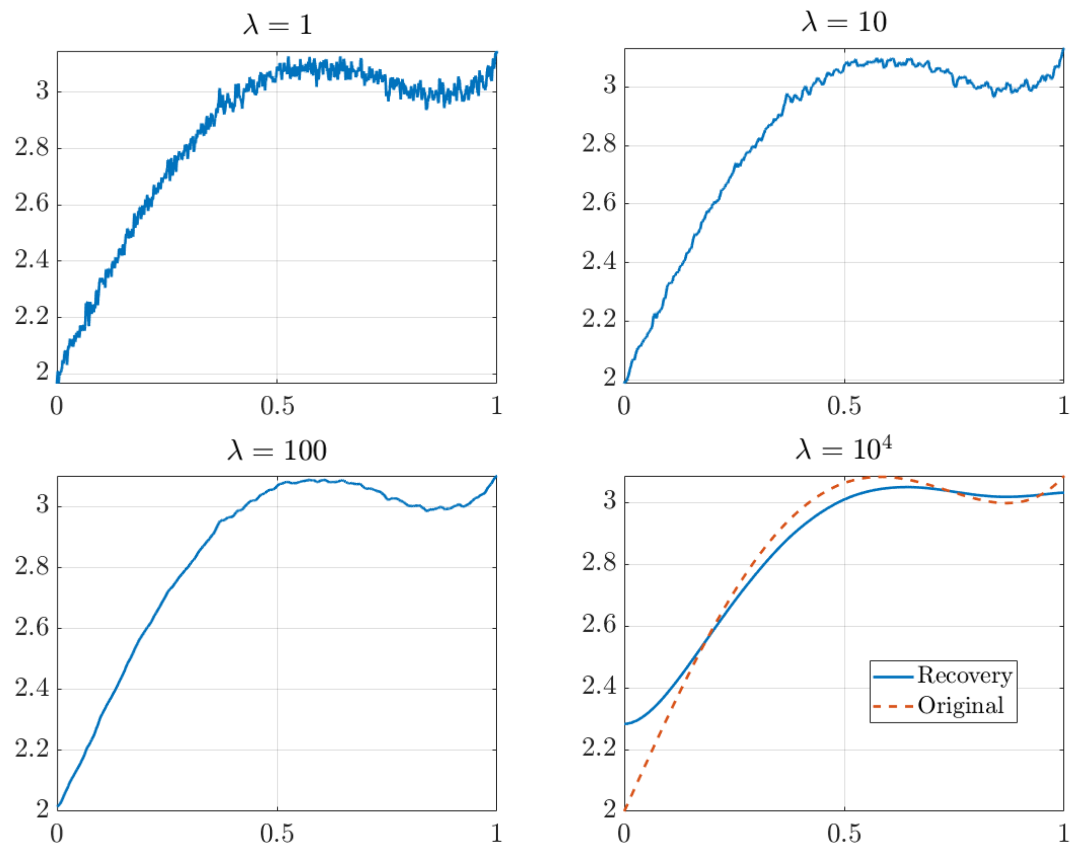

Figure 6: Signal denoising for different regularization parameters. Increasing the regularization parameters leads to better denoising, however, as the parameter becomes too large, the fit with the original signal is lost. What is the limit as λ grows?

As opposed to linear least squares, there is no easy way to to solve NLS problems. However, there are some dedicated algorithms for this problem, which we will explore later on in this module.

A Case Study: Circle Fitting

Given m points $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_m \in \mathbb{R}^n$, the circle fitting problem seeks to find a circle

$$C(\mathbf{x}, r) = \{\mathbf{y} \in \mathbb{R}^n : \|\mathbf{y} - \mathbf{x}\| = r\}$$

that best fits the m points.

We formulate a set of equations to match the center of the circle and its radius:

$$\|\mathbf{x} - \mathbf{a}_i\| \approx r, \quad i = 1, 2, \ldots, m\,.$$

To avoid nondifferentiability, consider the squared version:

$$\|\mathbf{x} - \mathbf{a}_i\|^2 \approx r^2, \quad i = 1, 2, \ldots, m\,.$$

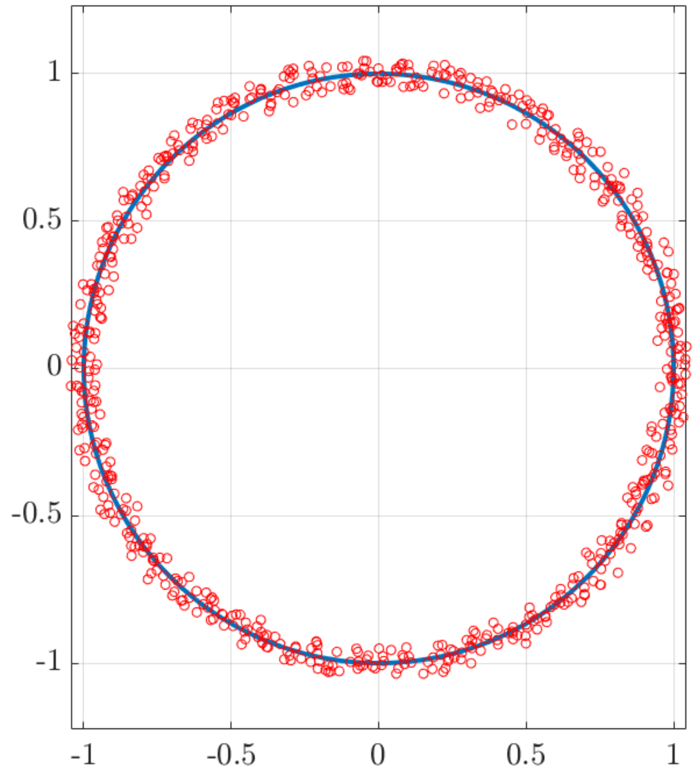

Figure 7: In the circle fitting problem, we look for a circle to match a set of measurements.

This leads to a Nonlinear Least Squares formulation:

$$\min_{\mathbf{x}\in\mathbb{R}^n,\ r\in\mathbb{R}_+} \sum_{i=1}^{m} \left(\|\mathbf{x}-\mathbf{a}_i\|^2 - r^2\right)^2,$$

or expanding

$$\min_{\mathbf{x},r} \left\{ \sum_{i=1}^{m} \left(-2\mathbf{a}_i^\top \mathbf{x} + \|\mathbf{x}\|^2 - r^2 + \|\mathbf{a}_i\|^2\right)^2 : \mathbf{x}\in\mathbb{R}^n, r\in\mathbb{R} \right\}$$

We will reduce this problem to a Linear Least Squares. Making the change of variables $R = \|x\|^2 - r^2$, the above problem reduces to

$$\min_{\mathbf{x}\in\mathbb{R}^n, R\in\mathbb{R}} \left\{ f(\mathbf{x},R) \equiv \sum_{i=1}^{m} \left(-2\mathbf{a}_i^\top \mathbf{x} + R + \|\mathbf{a}_i\|^2\right)^2 : \|\mathbf{x}\|^2 \geq R \right\}$$

The constraint $\|\mathbf{x}\|^2 \geq R$ can be dropped, and therefore the problem is equivalent to the LS problem

$$\min_{\mathbf{x},R} \left\{ \sum_{i=1}^{m} \left(-2\mathbf{a}_i^\top \mathbf{x} + R + \|\mathbf{a}_i\|^2\right)^2 : \mathbf{x}\in\mathbb{R}^n, R\in\mathbb{R} \right\}. \tag{CF-LS}$$

Redundancy of the Constraint $\|\mathbf{x}\|^2 \geq R$. We will show that any optimal solution $(\hat{\mathbf{x}}, \hat{R})$ of eq. (CF-LS) automatically satisfies $\|\hat{\mathbf{x}}\|^2 \geq \hat{R}$. Otherwise, if $\|\hat{\mathbf{x}}\|^2 < \hat{R}$, then

$$-2\mathbf{a}_i^\top \hat{\mathbf{x}} + \hat{R} + \|\mathbf{a}_i\|^2 > -2\mathbf{a}_i^\top \hat{\mathbf{x}} + \|\hat{\mathbf{x}}\|^2 + \|\mathbf{a}_i\|^2 = \|\hat{\mathbf{x}} - \mathbf{a}_i\|^2 \geq 0, \;\; i = 1, \ldots, m,$$

thus contradicting the optimality of $(\hat{\mathbf{x}}, \hat{R})$.